



Add Two Numbers Using Bit Manipulation

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You are given two integers A and B. Your task is to calculate their sum without using the + or - operators.

Use only bitwise operators to compute the result.

Input Format

A B

Constraints

$-10^9 \leq A, B \leq 10^9$

Output Format

Print a single integer representing the sum.

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Language

Python 3



```
1  import sys
2
3  def get_sum(a: int, b: int) -> int:
4
5      MASK = 0xFFFFFFFF
6      MAX_INT = 0x7FFFFFFF
7
8      while b != 0:
9
10         sum_no_carry = (a ^ b) & MASK
11         carry = ((a & b) << 1) & MASK
12         a = sum_no_carry
13         b = carry
14
15     return a if a <= MAX_INT else ~(a ^ MASK)
16
17 def solve():
18     input_data = sys.stdin.read().split()
19     if not input_data:
20         return
21     a = int(input_data[0])
22     b = int(input_data[1])
23     print(get_sum(a, b))
24
25 if __name__ == '__main__':
26     solve()
```

Line: 15 Col: 18

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5 7



* Trailing spaces detected in custom input

Compilation Successful :)

Click the Submit Code button to run your code against all the test cases.

Input (stdin)

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1	5 7
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Your Output (stdout)

1	12
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